

Swaroop Mudholkar

Electronics & Telecommunication Engineering Student | Embedded Systems | Automotive Electronics | AI & Computer Vision

✉ swaroopmudholkar@gmail.com ☎ +91 8830157488 📍 Pune, India

🌐 [linkedin.com/in/swaroop-mudholkar-entc6031a](https://www.linkedin.com/in/swaroop-mudholkar-entc6031a)

🔗 swaroop-mudholkar-portfolio.vercel.app 🐙 github.com/swaroop6031



EDUCATION

B.Tech in Electronics & Telecommunication

2023 – 2027 | Pune

Pimpri Chinchwad College of Engineering

12th Science, Maharashtra Board

2023 | Pune

S. B. Patil College of Science and Commerce

10th, CBSE

2021 | Pune

S. B. Patil Public School

SKILLS

Programming

C, Embedded C, C++, Python, CAPL, HTML, CSS, JavaScript

Embedded systems, Automotive & AI

STM32, ESP32, Arduino, CAN Bus, Vector CANoe, CANdb++, YOLOv8, OpenCV

Software Tools

MATLAB, Git, E-Plan Electric P8, Multisim

PROFESSIONAL EXPERIENCE

Cotmac Electronics Pvt. Ltd.

June 2025 – July 2025 | Pune

PLC Panel Design and Testing Intern

- Worked on PLC panel design, wiring verification, and component layout checks for industrial automation projects.
- Gained proficiency in E-Plan Electric P8, electrical schematic reading, and PLC-based control workflows.

PCCOE EnTC Internship Cell, Pune

Documentation Team Lead

- Led documentation, reporting, and team coordination for the department internship cell.

Technical Competitions

- Participated in Smart India Hackathon (SIH) 2024, IETE National Level Project Competition, and E-Yantra 2025.

PROJECTS

Proactive Detection of Aspergillus Growth Through Indoor Air Monitoring

Tech: STM32, UART, DHT22, MQ135, LCD

- Developed an embedded air-quality monitoring system for early fungal growth detection with real-time sensor monitoring and LCD-based alert indications.
- Published as a conference paper at the International Conference on Interdisciplinary Technology & Science Convergence (2025).

Engine ECU Simulation & CAN Network Validation

Tech: Vector CANoe, CAPL, CANdb++

- Developed multi-ECU CAN bus simulation with real-time speed monitoring, warning management, DTC handling, and bidirectional ECU communication.
- Implemented CAPL-based EngineECU and DashboardECU interaction with live dashboard visualization, CAN logging, and signal analysis.

Real-Time Object Detection using YOLOv7 & YOLOv8

Tech: Python, CNN, OpenCV, PyTorch

- Developed and compared YOLOv7 and YOLOv8 models for real-time object detection and performance analysis on edge-oriented applications.
- Evaluated detection accuracy, inference speed, and deployment feasibility using computer vision workflows.

Smart Chair System for Posture Monitoring & Wellness Reminder

Tech: ESP32, Force Resistor Sensors, Ultrasonic & Touch Sensors

- Designed a posture-monitoring system using ESP32 and sensors to detect incorrect sitting posture and provide real-time wellness reminders.